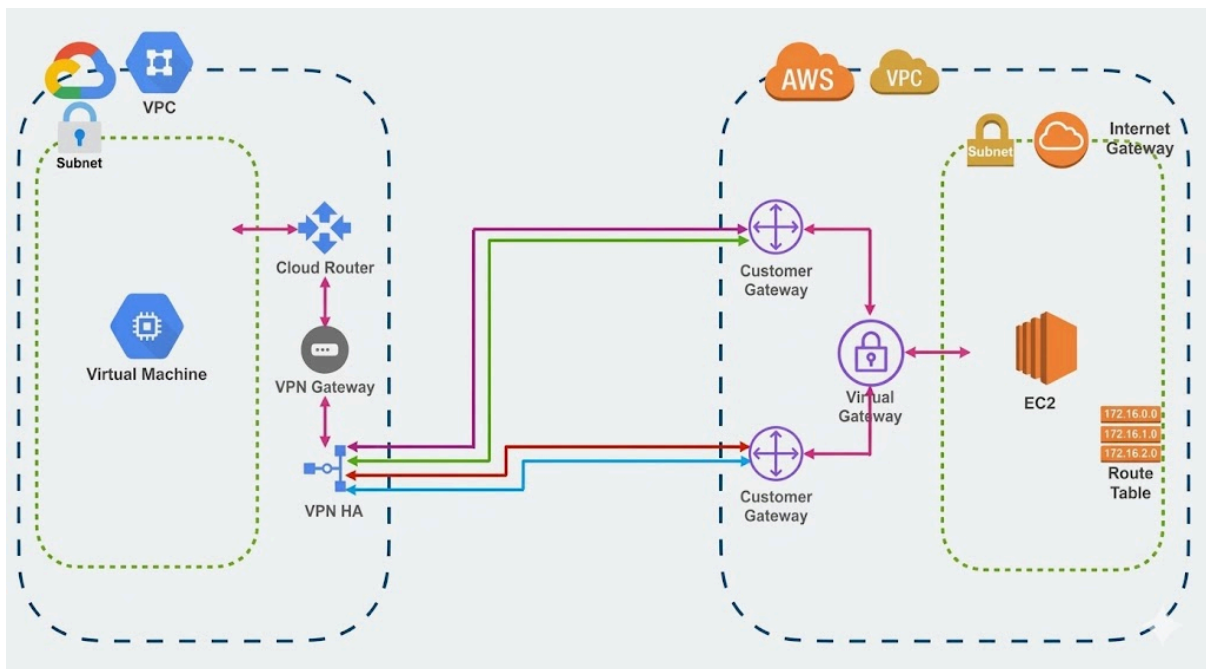


VPN Tunnel Documentation (AWS ↔ GCP – Complete Detailed Version)

The Architecture of VPN Tunnels:



Step 1: VPC Creation in AWS (vpc-mig)

Description:

This step shows the creation of a Virtual Private Cloud (VPC) in AWS, which serves as the base network infrastructure.

Action Performed:

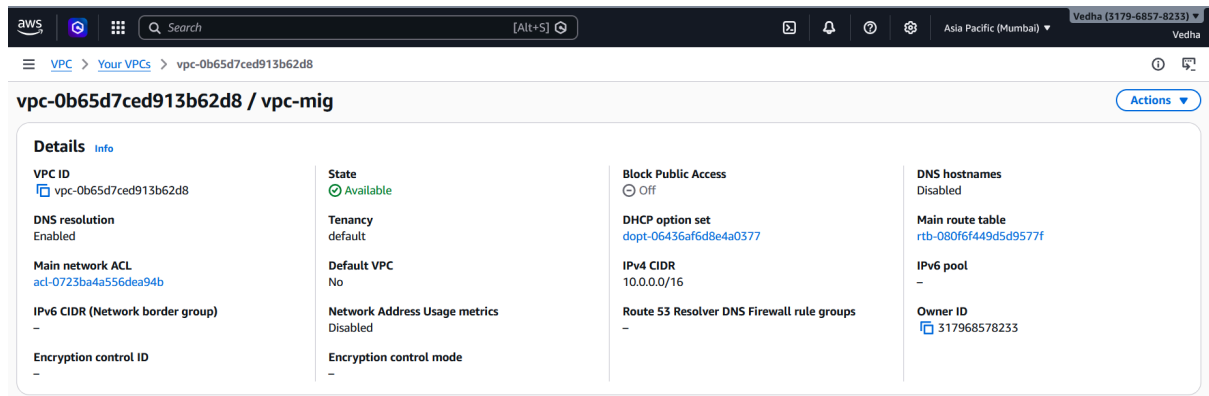
- VPC Name: **vpc-mig**
- VPC ID: **vpc-0b65d7ced913b62d8**
- IPv4 CIDR Block: **10.0.0.0/16**
- Region: **ap-south-1 (Mumbai)**
- Tenancy: Default

Purpose:

To create an isolated network environment in AWS for hosting resources.

Function:

The VPC provides IP addressing, routing, and network isolation required for establishing VPN connectivity with GCP.



Step 2: Subnet Creation in AWS (Public & Private)

Description:

Subnets are created to divide the VPC into smaller networks.

Action Performed:

Public Subnet:

- Name: **subnet-1-mig-public**
- Subnet ID: **subnet-017270a366520b356**
- CIDR: **10.0.1.0/24**
- Availability Zone: **ap-south-1a**

Private Subnet:

- Name: **subnet-2-mig-private**
- Subnet ID: **subnet-077868a2c28bcc757**
- CIDR: **10.0.2.0/24**
- Availability Zone: **ap-south-1b**

Purpose:

To logically separate resources into public and private segments.

Function:

Subnets control IP allocation and help isolate workloads for security and routing purposes.

aws

Search

[Alt+S]

Asia Pacific (Mumbai)

Vedha [3179-6857-8233]

Vedha

VPC

Subnets

subnet-017270a366520b356

subnet-017270a366520b356 / subnet-1-mig-public

Actions

Details

Subnet ID

subnet-017270a366520b356

IPv4 CIDR

10.0.1.0/24

Availability Zone

aps1-az1 (ap-south-1a)

Network ACL

acl-0723ba4a556dea94b

Auto-assign customer-owned IPv4 address

No

IPv6 CIDR reservations

-

Resource name DNS AAAA record

Disabled

Subnet ARN

arn:aws:ec2:ap-south-1:317968578233:subnet/subnet-017270a366520b356

Available IPv4 addresses

251

Network border group

ap-south-1

Default subnet

No

Customer-owned IPv4 pool

-

IPv6-only

No

DNS64

Disabled

State

Available

IPv6 CIDR

-

VPC

vpc-0b65d7ced913b62d8 | vpc-mig

Auto-assign public IPv4 address

No

Outpost ID

-

Hostname type

IP name

Owner

317968578233

Block Public Access

Off

IPv6 CIDR association ID

-

Route table

rtb-03eb131d16752ad6d | vpc-mig-public-routetable

Auto-assign IPv6 address

No

IPv4 CIDR reservations

-

Resource name DNS A record

Disabled

subnet-077868a2c28bcc757 / subnet-2-mig-private

Actions

Details

Subnet ID

subnet-077868a2c28bcc757

IPv4 CIDR

10.0.2.0/24

Availability Zone

aps1-az3 (ap-south-1b)

Network ACL

acl-0723ba4a556dea94b

Auto-assign customer-owned IPv4 address

No

IPv6 CIDR reservations

-

Resource name DNS AAAA record

Disabled

Subnet ARN

arn:aws:ec2:ap-south-1:317968578233:subnet/subnet-077868a2c28bcc757

Available IPv4 addresses

251

Network border group

ap-south-1

Default subnet

No

Customer-owned IPv4 pool

-

IPv6-only

No

DNS64

Disabled

State

Available

IPv6 CIDR

-

VPC

vpc-0b65d7ced913b62d8 | vpc-mig

Auto-assign public IPv4 address

No

Outpost ID

-

Hostname type

IP name

Owner

317968578233

Block Public Access

Off

IPv6 CIDR association ID

-

Route table

rtb-0cd014e72c10d1f3e | vpc-mig-private-routetable

Auto-assign IPv6 address

No

IPv4 CIDR reservations

-

Resource name DNS A record

Disabled

Step 3: Route Table Creation in AWS

Description:

Route tables define how traffic flows within the VPC.

Action Performed:

Public Route Table:

- Name: **vpc-mig-public-routetable**
- Routes:
 - 10.0.0.0/16 → local
 - 0.0.0.0/0 → Internet Gateway

Private Route Table:

- Name: **vpc-mig-private-routetable**
- Routes:
 - 10.0.0.0/16 → local

Purpose:

To control network traffic routing.

Function:

Route tables determine where packets are forwarded (internet, local, or VPN).

rtb-03eb131d16752ad6d / vpc-mig-public-routetable

Actions

Details

Info

Route table ID

rtb-03eb131d16752ad6d

VPC

vpc-0b65d7ced913b62d8 | vpc-mig

Main

No

Owner ID

317968578233

Explicit subnet associations

subnet-017270a366520b356 / subnet-1-mig-public

Edge associations

-

Routes

Subnet associations

Edge associations

Route propagation

Tags

Routes (4)

Both

Edit routes

Filter routes

Destination	Target	Status	Propagated	Route Origin
0.0.0.0/0	igw-0658967f1e8837f89	Active	No	Create Route
10.0.0.0/16	local	Active	No	Create Route Table
21.0.1.0/24	vgw-0e2f06aec1cd96c1a	Active	Yes	Virtual Private Gateway
21.0.2.0/24	vgw-0e2f06aec1cd96c1a	Active	Yes	Virtual Private Gateway

rtb-0cd014e72c10d1f3e / vpc-mig-private-routetable

Actions

Details Info

Route table ID

rtb-0cd014e72c10d1f3e

VPC

vpc-0b65d7ced913b62d8 | vpc-mig

Main

No

Owner ID

317968578233

Explicit subnet associations

subnet-077868a2c28bcc757 / subnet-2-mig-private

Edge associations

-

Routes

Subnet associationsEdge associationsRoute propagationTags

Routes (3)

BothEdit routes

Filter routes

Destination	Target	Status	Propagated	Route Origin
10.0.0.0/16	local	Active	No	Create Route Table
21.0.1.0/24	vgw-0e2f06aec1cd96c1a	Active	Yes	Virtual Private Gateway
21.0.2.0/24	vgw-0e2f06aec1cd96c1a	Active	Yes	Virtual Private Gateway

Step 4: Internet Gateway Creation and Attachment

Description:

Internet Gateway enables internet connectivity for AWS resources.

Action Performed:

- IGW Name: **vpc-mig-igw**
- Attached to: **vpc-mig**

Purpose:

To allow internet access for public subnet resources.

Function:

Acts as a bridge between AWS VPC and the internet.

igw-0658967f1e8837f89

Actions

Details Info

Internet gateway ID

igw-0658967f1e8837f89

State

Attached

VPC ID

vpc-0b65d7ced913b62d8 | vpc-mig

Owner

317968578233

Tags (1)

Manage tags

Search tags

Key	Value
Name	vpc-mig-internet-gateway

Step 5: VPC Creation in GCP

Description:

Creation of a custom VPC network in GCP.

Action Performed:

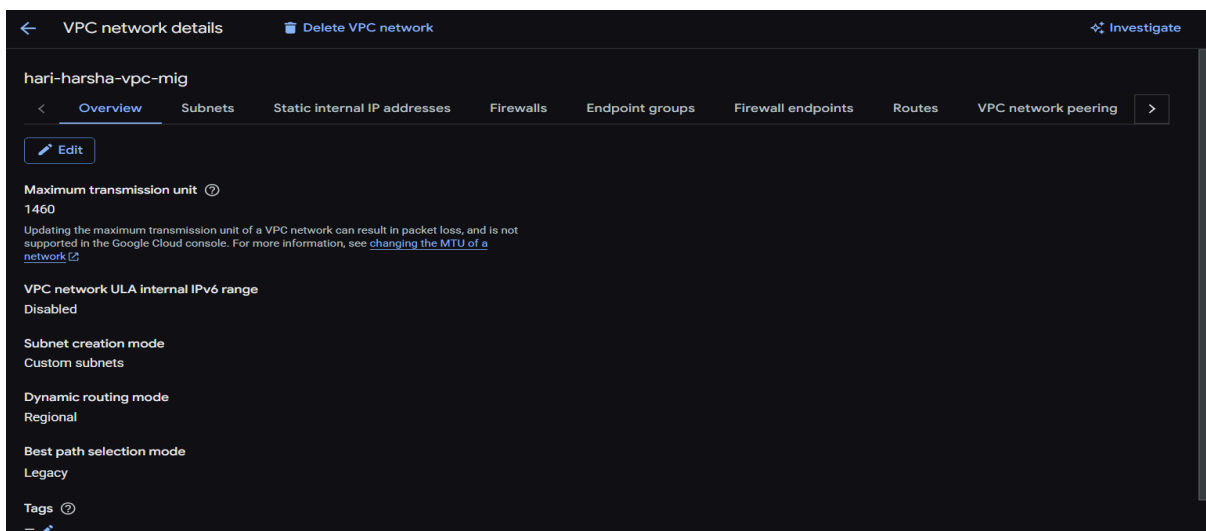
- VPC Name: *hari-harsha-vpc-mig*
- Mode: Custom
- Routing Mode: Regional

Purpose:

To create a GCP network environment.

Function:

Provides private networking and routing capabilities within GCP.



Step 6: Subnet Creation in GCP

Description:

A subnet is created inside the GCP VPC.

Action Performed:

- Subnet Name: (*subnet-private-mig, subnet-public-mig*)
- Region: (*asia-south1, mumbai*)
- CIDR Range: (*21.0.2.0/24, 21.0.1.0/24*)

Purpose:

To allocate IP ranges for GCP resources.

Function:

Defines IP addressing and regional placement.

← VPC network details		Delete VPC network		Investigate		Show info panel	
hari-harsha-vpc-mig							
< Overview Subnets Static internal IP addresses Firewalls Endpoint groups Firewall endpoints Routes VPC network peering >							
Subnets Add subnet Manage flow logs							
Filter Enter property name or value							
☐	Name ↑	Region	Stack type	Primary IPv4 range	Secondary IPv4 ranges	IPv6 ranges	Reserved internal range
☐	subnet-private-mig	asia-south1	IPv4 (single-stack)	21.0.2.0/24			None
☐	subnet-public-mig	asia-south1	IPv4 (single-stack)	21.0.1.0/24			None
Reserved proxy-only subnets for load balancing							
☐	Name	Region ↑	IP address ranges	Gateway	Role	Purpose	
No rows to display							
Equivalent REST							

Step 7: Cloud Router Creation (GCP)

Description:

Cloud Router is configured for dynamic routing using BGP.

Action Performed:

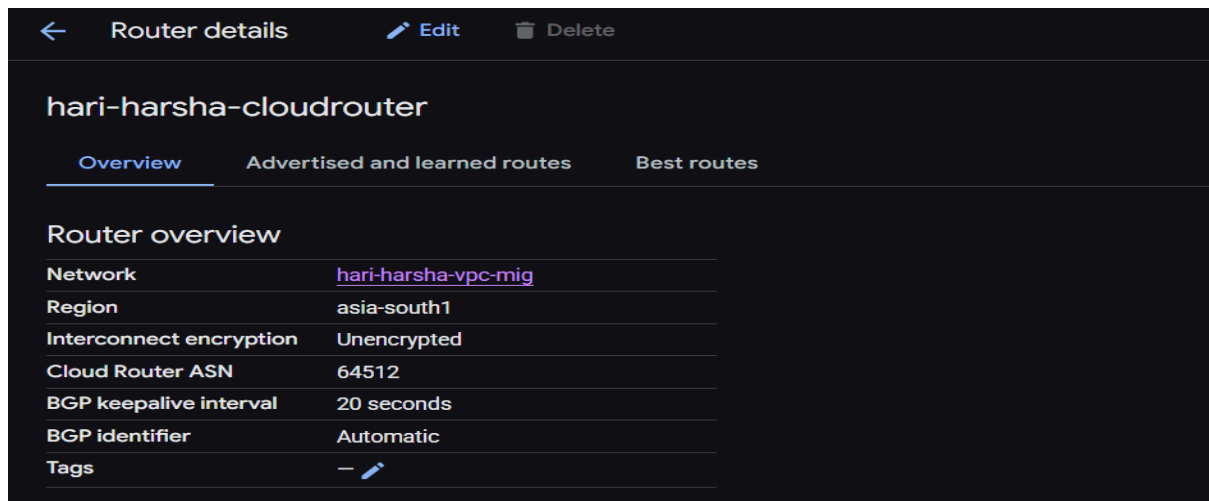
- Router Name: (*hari-harsha-cloudrouter*)
- Network: hari-harsha-vpc-mig
- Region: asia-south1
- ASN: **64512**

Purpose:

To enable dynamic route exchange.

Function:

Cloud Router uses BGP to exchange routes between AWS and GCP.



Step 8: HA VPN Gateway Creation (GCP)

Description:

HA VPN gateway is created for high availability.

Action Performed:

- VPN Name: *hari-harsah-vpn-gateway*
- Network: hari-harsha-vpc-mig
- Region: asia-south1

Generated IPs:

- Interface 0 IP: (*Public IP1: 34.152.107.255*)
- Interface 1 IP: (*Public IP2: 34.177.75.42*)

Purpose:

To provide secure VPN endpoints.

Function:

HA VPN ensures redundancy with two tunnels.

VPN									
VPN setup wizard Refresh Recommended alerts Show info panel									
Cloud VPN tunnels Cloud VPN gateways Peer VPN gateways Observability									
Create VPN gateway									
Filter Enter property name or value									
☐	Gateway name ↑	IP version	IP address	VPC network	Region	VPN tunnels	Description	Labels	Actions
☐	hari-harsha-vpn-gateway	IPv4	Interface: 0 34.152.107.255 Interface: 1 34.177.75.42	hari-harsha-vpc-mig	asia-south1	tunnel-1-hari-harsha, tunnel-2-hari-harsha, tunnel-3-hari-harsha, tunnel-4-hari-harsha			Add VPN tunnel ⋮

Step 9: Copy HA VPN External IP

Description:

Public IP of GCP VPN is copied.

Action Performed:

- Selected Interface 0 IP: (34.152.107.255)

Purpose:

Used for AWS configuration.

Function:

Acts as a remote endpoint IP.

Step 10: Customer Gateway Creation (AWS)

Description:

Defines GCP as a remote VPN endpoint in AWS.

Action Performed:

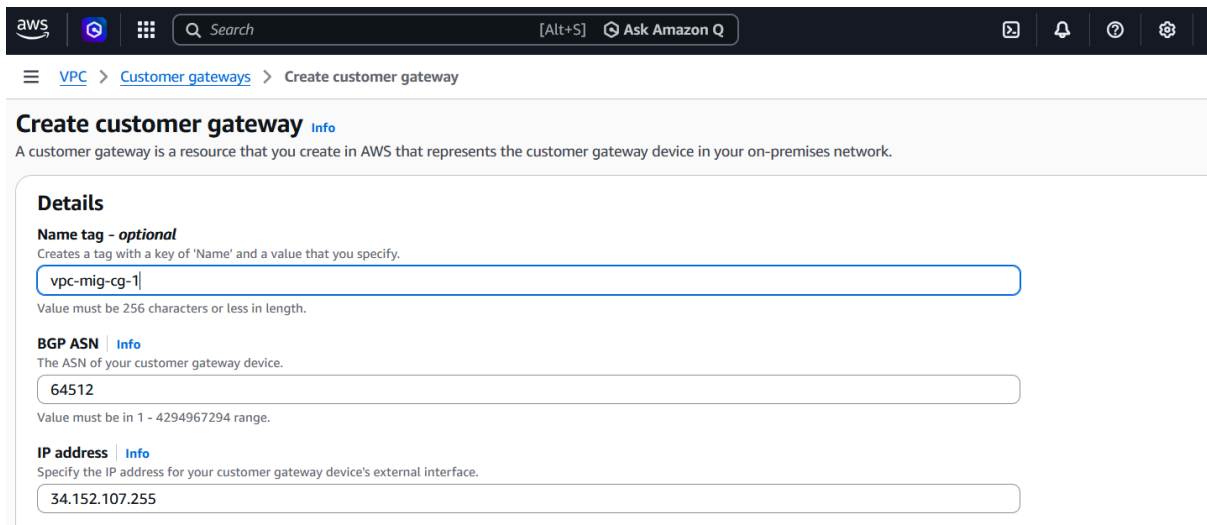
- Name: **vpc-mig-cg-1**
- IP Address: (34.152.107.255)
- Routing: Dynamic (BGP)
- BGP ASN: **64512**

Purpose:

To represent GCP in AWS.

Function:

Customer Gateway stores remote peer details.



The screenshot shows the AWS Management Console interface for creating a customer gateway. The breadcrumb navigation at the top reads: VPC > Customer gateways > Create customer gateway. The main heading is 'Create customer gateway' with an 'Info' link. Below the heading is a descriptive sentence: 'A customer gateway is a resource that you create in AWS that represents the customer gateway device in your on-premises network.' The 'Details' section contains three input fields: 1. 'Name tag - optional' with a description 'Creates a tag with a key of 'Name' and a value that you specify.' and a value of 'vpc-mig-cg-1'. 2. 'BGP ASN' with a description 'The ASN of your customer gateway device.' and a value of '64512'. 3. 'IP address' with a description 'Specify the IP address for your customer gateway device's external interface.' and a value of '34.152.107.255'.

aws [Search] [Alt+S] Ask Amazon Q

VPC > Customer gateways > Create customer gateway

Create customer gateway [Info](#)

A customer gateway is a resource that you create in AWS that represents the customer gateway device in your on-premises network.

Details

Name tag - optional
Creates a tag with a key of 'Name' and a value that you specify.

vpc-mig-cg-1

Value must be 256 characters or less in length.

BGP ASN [Info](#)
The ASN of your customer gateway device.

64512

Value must be in 1 - 4294967294 range.

IP address [Info](#)
Specify the IP address for your customer gateway device's external interface.

34.152.107.255

Step 11: Virtual Private Gateway Creation (AWS)

Description:

VGW acts as AWS VPN endpoint.

Action Performed:

- Name: **vpc-mig-vpg**
- ASN: **64513**

Purpose:

To terminate VPN connection in AWS.

Function:

Handles IPSec tunnels and routing.

vgw-0e2f06aec1cd96c1a / vpc-mig-vpg

info

Actions

Details

Virtual private gateway ID

vgw-0e2f06aec1cd96c1a

Amazon ASN

64513

State

Available

VPC attachment state

Attached

Type

ipsec.1

VPC

vpc-0b65d7ced913b62d8 | vpc-mig

Tags

Tags (1)

Manage tags

Search tags

Key

Value

Name

vpc-mig-vpg

Step 12: Attach VGW to VPC

Description:

VGW is attached to AWS VPC.

Action Performed:

- VGW: vpc-mig-vpg
- VPC: vpc-mig

Purpose:

To enable VPN traffic into VPC.

Function:

Links VPN gateway with AWS network.

Step 13: Enable Route Propagation

Description:

Allows automatic route updates.

Action Performed:

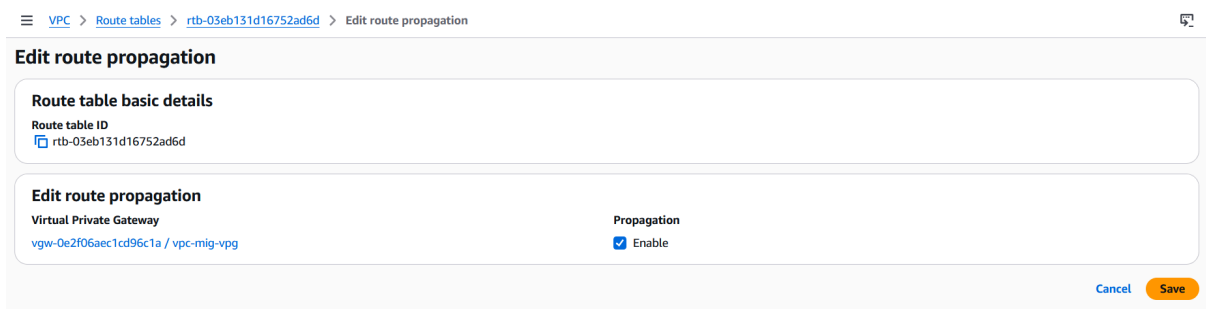
- Route Table: vpc-mig-private-routetable and vpc-mig-public-routetable
- Propagation: Enabled

Purpose:

To automatically learn VPN routes.

Function:

BGP routes are added dynamically.



The screenshot shows the 'Edit route propagation' page in the AWS Management Console. The breadcrumb navigation at the top reads: VPC > Route tables > rtb-03eb131d16752ad6d > Edit route propagation. The page title is 'Edit route propagation'. There are two main sections: 'Route table basic details' and 'Edit route propagation'. The 'Route table basic details' section shows the 'Route table ID' as 'rtb-03eb131d16752ad6d'. The 'Edit route propagation' section shows the 'Virtual Private Gateway' as 'vgw-0e2f06aec1cd96c1a / vpc-mig-vpg' and the 'Propagation' status as 'Enable' with a checked checkbox. At the bottom right, there are 'Cancel' and 'Save' buttons.

Step 14: Site-to-Site VPN Creation (AWS)

Description:

Creates VPN connection between AWS and GCP.

Action Performed:

- VPN Name: **vpc-mig**
- VGW: vpc-mig-vpg
- Customer Gateway: vpc-mig-cg-01 and vpc-mig-cg-02
- Routing: Dynamic

Purpose:

To establish IPsec connection.

Function:

Creates encrypted tunnels.

Create VPN connection [info](#)

Select the resources and additional configuration options that you want to use for the site-to-site VPN connection.

Details

Name tag - optional

Creates a tag with a key of 'Name' and a value that you specify.

vpn-mig-01

Value must be 256 characters or less in length.

Target gateway type [info](#)

☒ Virtual private gateway

☐ Transit gateway

☐ Site-to-Site VPN Concentrator

☐ Not associated

Virtual private gateway

vgw-0e2f06aec1cd96c1a

Customer gateway [info](#)

☒ Existing

☐ New

Customer gateway ID

cgw-08517f80473a1eb44

Step 15: Download VPN Configuration File

Description:

Configuration file downloaded.

Action Performed:

- Vendor: Generic

Purpose:

To get tunnel and BGP details.

Function:

Contains all required parameters.

vpn-0ca4eba023489082c / vpc-mig-vpn-01 [info](#)

Download configuration

Actions

Details

VPN ID

vpn-0ca4eba023489082c

Transit gateway

-

VPC

-

Local IPv4 network CIDR

0.0.0.0/0

Core network ARN

-

Secrets management ARN

-

State

Available

Customer gateway address

34.152.107.255

Routing

Dynamic

Remote IPv4 network CIDR

0.0.0.0/0

Core network attachment ARN

-

Tunnel bandwidth

Standard

Virtual private gateway

vgw-0e2f06aec1cd96c1a

Type

ipsec.1

Acceleration enabled

Disabled

Local IPv6 network CIDR

-

Gateway association state

Associated

VPN Concentrator ID

-

Customer gateway

cgw-08517f80473a1eb44

Category

VPN

Authentication

Pre-shared key

Remote IPv6 network CIDR

-

Outside IP address type

Public IPv4

Download configuration

×

Download a sample configuration based on your customer gateway. Note that this is a sample only, and that it will require modification for using Advanced Algorithms, Certificates, and/IPv6.

Vendor
The manufacturer of the customer gateway device (for example, Cisco Systems, Inc).

Generic

Platform
The class of the customer gateway device (for example, J-Series).

Generic

Software
The operating system running on the customer gateway device (for example, ScreenOS).

Vendor Agnostic

IKE version
The IKE version you are using for your VPN connection.

ikev2

Include sample type - optional
☐ Enable

Sample type
The default sample type compatibility mode includes all options. The recommended mode restricts

CancelDownload

Step 16: Extract Values from Config File

Description:

Important values extracted.

Extracted:

- Tunnel IPs
- BGP IPs
- ASN
- Pre-shared Keys

Purpose:

Used in GCP setup.

Function:

Provides configuration details.

Step 17: GCP Tunnel & BGP Configuration

Action Performed:

Tunnel 1:

- Name: tunnel-1
- Peer IP: 3.108.91.188
- Shared Key: kXPj_xRbQLVcLBKoMwUx9RBet.LeFCMy

BGP:

- Peer ASN: 64512
- BGP IPs: 169.254.74.178 and 169.254.74.177

Tunnel 2:

- Name: tunnel-2
- Peer IP: 13.232.98.171
- Shared Key: DMEemVGe7YwZnCK_fkQsIII9FIaFKTD9

BGP:

- Peer ASN: 64512
- BGP IPs: 169.254.125.14 and 169.254.125.13

Tunnel 3:

- Name: tunnel-3
- Peer IP: 3.7.175.245
- Shared Key: k57QAoaD.nK5I5HvsEXPwImGGvuTbwim

BGP:

- Peer ASN: 64512
- BGP IPs: 169.254.28.238 and 169.254.28.237

Tunnel 4:

- Name: tunnel-4
- Peer IP: 13.206.49.249

- Shared Key: bps87XveuATsrN8mVvwkn3j8Feac3_F4

BGP:

- Peer ASN: 64512
- BGP IPs: 169.254.202.210 and 169.254.202.209

Purpose:

To establish redundancy.

Function:

Creates two tunnels with failover.

```
- IKE version           : IKEv2
- Authentication Method : Pre-Shared Key
- Pre-Shared Key        : kXPj_xRbQLVcLBKoMwUx9RBEt.LeFCMy
- Authentication Algorithm : sha1
- Encryption Algorithm  : aes-128-cbc
- Lifetime              : 28800 seconds
- Phase 1 Negotiation Mode : main
- Diffie-Hellman        : Group 2
```

Outside IP Addresses:

```
- Customer Gateway      : 34.152.107.255
- Virtual Private Gateway : 3.108.91.188
```

Inside IP Addresses

```
- Customer Gateway      : 169.254.74.178/30
- Virtual Private Gateway : 169.254.74.177/30
```


Edit BGP session

IPv4 BGP session

Name *
tunnel-1-bgp-1-h

Lowercase letters, numbers, hyphens allowed

Peer ASN *
64513

Advertised route priority (MED)
MED value is used for active/passive configuration

Multiprotocol BGP

BGP sessions are set up over IPv4, exchanging IPv4 and IPv6 addresses

☐ Enable IPv6 traffic

You need to enable IPv6 traffic to allocate BGP next hops for IPv6 traffic.

Allocate BGP IPv4 address

☐ Automatically

☒ Manually

Cloud Router BGP IPv4 address *
169.254.74.178

BGP peer IPv4 address *
169.254.74.177

Step 18: Create Testing Instances (GCP VM & AWS EC2)

Action Performed:

GCP: Created VM instance **hari-harsha-vm-testing**

- Network: hari-harsha-vpc-mig
- Subnet: subnet-public-mig
- Internal IP: **21.0.1.2**
- External IP: **34.93.2.1**
- Zone: asia-south1-c

AWS: Created EC2 instance **testing-private-ec2**

- Instance ID: **i-03cf2e086cad283ff**
- VPC: vpc-mig
- Subnet: subnet-2-mig-private
- Private IP: **10.0.2.25**
- Public IP: **Not assigned**
- Instance State: Running

Purpose:

To set up source and destination instances in GCP and AWS for testing VPN connectivity using private IP communication.

Function:

GCP VM (public + private IP) acts as source, and AWS EC2 (private-only) acts as destination, ensuring that communication happens securely over the VPN tunnel without internet exposure.

Basic information

Name	hari-harsha-vm-testing
Instance ID	3816326041849293026
Description	None
Type	Instance
Status	Running

Network interfaces

Filter

Enter property name or value

Name	Network	Subnetwork	NIC type	Primary internal IP address	Alias IP ranges	IP stack type	External IP
nic0	hari-harsha-vpc-mig	subnet-public-mig		21.0.1.2		IPv4	34.93.2.1 (Eg

:f2e086cad283ff

Instance summary for i-03cf2e086cad283ff (testing-private-ec2) Info

Connect

Instance state

Actions

Updated 1 minute ago

Instance ID

i-03cf2e086cad283ff

IPv6 address

-

Hostname type

IP name: ip-10-0-2-25.ap-south-1.compute.internal

Answer private resource DNS name

-

Auto-assigned IP address

-

IAM role

-

Public IPv4 address

-

Instance state

Running

Private IP DNS name (IPv4 only)

ip-10-0-2-25.ap-south-1.compute.internal

Instance type

t2.micro

VPC ID

vpc-0b65d7ced913b62d8 (vpc-mig)

Subnet ID

subnet-077868a2c28bcc757 (subnet-2-mig-private)

Private IPv4 addresses

10.0.2.25

Public DNS

-

Elastic IP addresses

-

AWS Compute Optimizer finding

Opt-in to AWS Compute Optimizer for recommendations. | Learn more

Auto Scaling Group name

-

Step 19: Verification & Testing

Action Performed:

- GCP: Tunnel status = Established
- AWS: Tunnel status = UP
- Routing verified
- Ping test successful

Purpose:

To confirm connectivity.

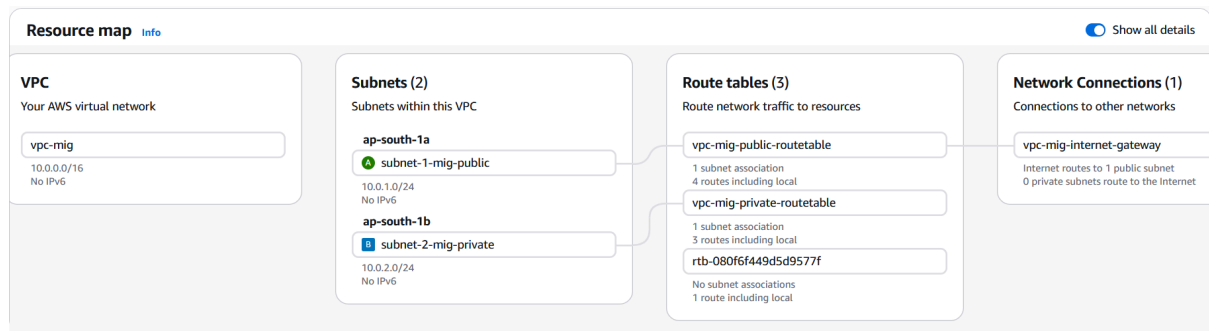
Function:

Ensures VPN is working end-to-end.

VPN							
VPN setup wizard Refresh Recommended alerts Show info panel							
Cloud VPN tunnels Cloud VPN gateways Peer VPN gateways Observability							
Create VPN tunnel							
VPN tunnels Delete Manage flow logs							
Filter Enter property name or value							
	Name	Cloud VPN gateway (IP)	Peer VPN gateway (IP)	Cloud Router BGP IP address	Peer BGP IP address	VPN tunnel status	Actions
	tunnel-1-hari-harsha	hari-harsha-vpn-gateway34.152.107.255	hari-harsha-vpng-mig3.108.91.188	169.254.74.178	169.254.74.177	Established	
	tunnel-2-hari-harsha	hari-harsha-vpn-gateway34.152.107.255	hari-harsha-vpng-mig13.232.98.171	169.254.125.14	169.254.125.13	Established	
	tunnel-3-hari-harsha	hari-harsha-vpn-gateway34.177.75.42	hari-harsha-vpng-mig3.7.175.245	169.254.28.238	169.254.28.237	Established	
	tunnel-4-hari-harsha	hari-harsha-vpn-gateway34.177.75.42	hari-harsha-vpng-mig13.206.49.249	169.254.202.210	169.254.202.209	Established	

Tunnel details Tags								
Tunnel state								
Tunnel number	Outside IP address	Inside IPv4 CIDR	Inside IPv6 CIDR	Status	Provisioning status	Last status change	Details	Certificate ARN
Tunnel 1	3.108.91.188	169.254.74.176/30	–	Up	Available	April 5, 2026, 12:10:11 (UTC+05:30)	2 BGP ROUTES	–
Tunnel 2	13.232.98.171	169.254.125.12/30	–	Up	Available	April 5, 2026, 12:09:27 (UTC+05:30)	2 BGP ROUTES	–

The final AWS VPN Network Config:



Connectivity Test from GCP VM to AWS EC2 (Ping Test)

Description:

This screenshot shows the connectivity test performed from the GCP Virtual Machine to the AWS EC2 instance using the private IP address over the VPN tunnel.

Action Performed:

- Accessed GCP VM using SSH (hari-harsha-vm-testing)
- Executed ping command:

ping 10.0.2.25

- Target IP: 10.0.2.25 (AWS EC2 Private IP)

Observed Output:

- Replies received successfully from 10.0.2.25
- Packet transmission:
 - 31 packets transmitted
 - 31 packets received
 - 0% packet loss
- Latency:
 - Min: 4.024 ms
 - Avg: 4.515 ms
 - Max: 9.639 ms

Purpose:

To verify that the GCP VM can communicate with AWS EC2 over the private VPN tunnel.

Function:

This confirms that:

- VPN tunnel is active
- Routing is correctly configured
- BGP route exchange is successful
- No packet loss indicates stable connectivity

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hariharan_j_cloudambassadors_com@hari-harsha-vm-testing:~$ ping 10.0.2.25
PING 10.0.2.25 (10.0.2.25) 56(84) bytes of data:
64 bytes from 10.0.2.25: icmp_seq=1 ttl=63 time=9.64 ms
64 bytes from 10.0.2.25: icmp_seq=2 ttl=63 time=4.90 ms
64 bytes from 10.0.2.25: icmp_seq=3 ttl=63 time=5.11 ms
64 bytes from 10.0.2.25: icmp_seq=4 ttl=63 time=5.15 ms
64 bytes from 10.0.2.25: icmp_seq=5 ttl=63 time=4.09 ms
64 bytes from 10.0.2.25: icmp_seq=6 ttl=63 time=4.05 ms
64 bytes from 10.0.2.25: icmp_seq=7 ttl=63 time=4.29 ms
64 bytes from 10.0.2.25: icmp_seq=8 ttl=63 time=4.37 ms
64 bytes from 10.0.2.25: icmp_seq=9 ttl=63 time=4.43 ms
64 bytes from 10.0.2.25: icmp_seq=10 ttl=63 time=4.38 ms
64 bytes from 10.0.2.25: icmp_seq=11 ttl=63 time=4.50 ms
64 bytes from 10.0.2.25: icmp_seq=12 ttl=63 time=4.12 ms
64 bytes from 10.0.2.25: icmp_seq=13 ttl=63 time=4.12 ms
64 bytes from 10.0.2.25: icmp_seq=14 ttl=63 time=4.06 ms
64 bytes from 10.0.2.25: icmp_seq=15 ttl=63 time=4.29 ms
64 bytes from 10.0.2.25: icmp_seq=16 ttl=63 time=4.56 ms
64 bytes from 10.0.2.25: icmp_seq=17 ttl=63 time=4.43 ms
64 bytes from 10.0.2.25: icmp_seq=18 ttl=63 time=4.07 ms
64 bytes from 10.0.2.25: icmp_seq=19 ttl=63 time=4.11 ms
64 bytes from 10.0.2.25: icmp_seq=20 ttl=63 time=4.02 ms
64 bytes from 10.0.2.25: icmp_seq=21 ttl=63 time=4.29 ms
64 bytes from 10.0.2.25: icmp_seq=22 ttl=63 time=4.65 ms
64 bytes from 10.0.2.25: icmp_seq=23 ttl=63 time=4.77 ms
64 bytes from 10.0.2.25: icmp_seq=24 ttl=63 time=4.14 ms
64 bytes from 10.0.2.25: icmp_seq=25 ttl=63 time=4.04 ms
64 bytes from 10.0.2.25: icmp_seq=26 ttl=63 time=4.14 ms
64 bytes from 10.0.2.25: icmp_seq=27 ttl=63 time=4.16 ms
64 bytes from 10.0.2.25: icmp_seq=28 ttl=63 time=4.21 ms
64 bytes from 10.0.2.25: icmp_seq=29 ttl=63 time=4.45 ms
64 bytes from 10.0.2.25: icmp_seq=30 ttl=63 time=4.17 ms
64 bytes from 10.0.2.25: icmp_seq=31 ttl=63 time=4.26 ms
^C
--- 10.0.2.25 ping statistics ---
31 packets transmitted, 31 received, 0% packet loss, time 30041ms
rtt min/avg/max/mdev = 4.024/4.515/9.639/0.981 ms
hariharan_j_cloudambassadors_com@hari-harsha-vm-testing:~$
```

Final Outcome

A secure, highly available Site-to-Site VPN tunnel between AWS and GCP is successfully established using BGP dynamic routing.